

# EMICS'21: Eye Movements as an Interface to Cognitive State

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## ABSTRACT

Eye movement recording has been extensively used in HCI and offers the possibility to understand how information is perceived and processed by users. Hardware developments provide the ubiquitous accessibility of eye recording, allowing eye movements to enter common usage as a control modality. Recent A.I. developments provide powerful computational means to make predictions about the user. However, the connection between eye movements and cognitive state has been largely under-exploited in HCI. Despite the rich literature in psychology, a deeper understanding of its usability in practice is still required. This virtual EMICS workshop will provide an opportunity to discuss possible application scenarios and HCI interfaces to infer users' mental state from eye movements. It will bring together researchers across disciplines with the goal of expanding shared knowledge, discussing innovative research directions and methods, fostering future collaborations around the use of eye movements as an interface to cognitive state.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

## KEYWORDS

eye movements, cognitive states, attention, interface, interaction modality

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## 1 BACKGROUND

The ability to leverage eye tracking data to infer perceptual and cognitive processes during activities such as reading, driving and medical image examinations [2, 3, 13, 14, 21, 26], has long been

used in academic settings. However, as eye tracking systems become ubiquitous across different interfaces (e.g. desktop, mobile and head-mounted displays), new opportunities arise for applying this knowledge to human-computer interaction. This virtual EMICS workshop will be used to explore applications that can be made possible by connecting eye movements, cognitive processing, and cognitive state [6, 10, 16, 25].

The cognitive state of a user can inform not only whether or not they stay on task, but what type of thinking (explorative, expert, creative) is currently engaged [9]. With the advent of A.I. in this field, eye movement recordings can be used to decode, guide and encourage different types of cognitive processing [1, 5, 7, 8, 12, 15, 17, 23, 28, 32].

At the same time, the use of eye tracking in HCI has been highly promising for many years, but progress has been slow. In Jacob and Karn's review of eye tracking in HCI and usability research, they state, "We see promising research work, but we have not yet seen wide use of these approaches in practice or in the marketplace" [19] and it remains as true today as it was in 2003.

Following our CHI '20 EMICS '20 virtual SIG meeting [31], the proposed CHI '21 virtual workshop will be used as a space in which researchers from different disciplines (HCI, psychology, A.I., cognitive neuroscience) can interact and strengthen this budding field. Eye movements can, and have been, used as an input modality on mobile and head-mounted displays; for instance, for text entry or search [24, 27]. However, the main focus of this workshop is on how eye movement patterns can be used to evaluate user intentions and task difficulty, by providing a window into the user's cognitive processing and state. The use of eye movements to decode cognitive states could be extended to adaptive interfaces that use, for instance, eye fixations as feedback to guide attention [18, 22, 29], affect processing [20, 26], and provide aids for learning [30] and memory [4, 11]. The development of such tools could have some far reaching implications for our society and launch a new innovative approach to human-computer interaction.

### 1.1 Relevance of the workshop

Inclusion of eye trackers in a wide variety of mobile applications and head-mounted interfaces is growing in popularity (and affordability). Examples include the commercial off-the-shelf availability of the HTC Vive Pro Eye, Microsoft's HoloLens 2, Tobii Glasses 3, and Pupil Labs' *Invisible* wearable device – to name a few. With the

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increasing accessibility of eye tracking devices, open source software (e.g., OpenVR, Pupil Labs entire software suite, PyOpenGaze), and the development of powerful A.I. algorithms, inferring cognitive states from eye movement recordings has become feasible across many scenarios. These advancements open the possibility of integrating users' cognitive processing into the design of interactive systems. Many studies have been conducted in psychology to understand the correspondence between eye movements and cognitive processing, however this connection has not yet been fully explored or implemented in HCI. So far it remains unclear how to create meaningful interactions that make full use of the information revealed by eye movements. With the growing interest in the inference of users' cognitive states and processing, we see an increased demand for bringing together the Cognitive Science and the HCI communities to share knowledge and explore the full potential of this intersection of research interests and its applications.

## 1.2 Related initiatives

There are numerous initiatives related to eye tracking, e.g., workshops such as EMDPA (Special Session on Eye Movement Data Processing and Analysis), ETTAC (Workshop on Eye Tracking Techniques, Applications and Challenges), ETAVI (Workshop on Eye Tracking in Aviation), and EARLI SIG 27 (Online Measures of Learning Processes), to name a handful. These initiatives are generally related to a given domain, e.g., aviation, learning, or data processing (signal analysis), where eye tracking is specifically used to measure human visual attention within these domain-related tasks. Meanwhile, the ACM's flagship conference related to eye tracking, the Symposium on Eye Tracking Research & Applications (ETRA), is a related stand-alone initiative, bringing together computer scientists, engineers, and behavioral scientists. ETRA covers the gamut in research related to eye tracking, featuring work that is broadly related (hardware, visualizations, etc.). In contrast to other workshops or to ETRA, we constrain our EMICS workshop call to the specific confluence of eye movements and cognitive state, which will help to focus the submissions, applications, and discussions at the workshop. The effort to bring more Cognitive and Vision Scientists to CHI parallels a similar effort by the VisXVision team (<https://visxvision.com/>) to organize workshops and symposia at the intersection of information visualization and vision science, which has gained great visibility at the yearly VIS and VSS conferences, respectively. One of the EMICS organizers has been part of the VisXVision team, with experience organizing interdisciplinary workshops. In turn, expected outcomes derived from eye tracking and estimation of cognitive state will impact domain-specific research covered by related workshops, and will also impact general eye tracking research covered by ETRA.

## 1.3 Workshop aims and expected outcomes

The main objective of the virtual EMICS workshop is to bring together researchers and practitioners from academia and industry to explore the opportunities afforded by recording eye movement patterns to infer cognitive state and elevate human-computer interactions. We will have a set of prerecorded lightning talks from invited panelists kick-off an open discussion about the current

cutting-edge use cases of eye movements in HCI and cognitive psychology, as well as the potential challenges and future directions in this space. Specifically, the virtual workshop will address the following discussion points:

- Which cognitive states can be most reliably and robustly inferred from eye movements in practice?
- Which applications of eye movements as an interface to cognitive state (EMICS) have already proven successful?
- What challenges and limitations of current hardware, algorithms, etc. need to be addressed to facilitate future applications of EMICS?
- What are the design principles for such interface where eye movements are used to infer the mental states? What are the potential challenges and limitations?
- How can we best align the interests between academia and industry to solve these challenges?

## 1.4 Diversity

The above goals will be accomplished by inviting leaders from the Cognitive Science and HCI fields for an impressive speaker line-up, with 9 planned talks. Speakers represent academic labs across the US, Europe and Asia-Pacific. The organising committee is composed of 3 academics and 2 industrial research scientists, bringing a diverse mix of viewpoints to this first workshop on EMICS.

## 2 ORGANIZERS

**Xi Wang**, Ph.D., has recently graduated with distinction from the Technische Universität Berlin. She has an M.Eng. degree from Shanghai Jiao Tong University and an M.Sc. degree in computer science from Technische Universität Berlin. Her research is broadly concerned with human perception and its applications to computer graphics and human-computer interactions. She has worked on projects studying vergence of eye movements in 3D space, measuring where humans look on real three-dimensional stimuli, and using a mental imagery paradigm to retrieve the mental imagery from eye movements while looking at nothing.

**Zoya Bylinskii**, Ph.D., is a Research Scientist at Adobe Inc. She received a Ph.D. in Electrical Engineering and Computer Science from MIT in 2018 and an Hon. B.Sc. in Computer Science and Statistics from the University of Toronto in 2012. Zoya is a 2016 Adobe Research Fellow, a 2014-2016 NSERC Postgraduate Scholar, a 2013 Julie Payette Research Scholar, and a 2011 Anita Borg Scholar. Zoya works at the interface of human vision, computer vision, and human-computer interaction: building computational models of people's memory and attention, and applying the findings to graphic designs and data visualizations.

**Monica Castelhan**, Ph.D., is a Professor in Psychology at Queen's University, Canada. Dr. Castelhan studies memory and attentional processes in complex visual environments in real and virtual settings using techniques such as VR, eye tracking and electroencephalogram (EEG). Dr. Castelhan has been granted numerous awards including the Early Researcher Award from ORF and numerous grants from national and international funding agencies.

**James Hillis**, Ph.D., is a Research Scientist at Facebook Reality Labs. He completed a Vision Science Ph.D. in sensory cue combination at UC Berkeley in 2002 and post-doctoral research in color vision at the University of Pennsylvania before joining the faculty at the University of Glasgow in 2006. At the University of Glasgow he headed a lab studying socio-economic decision making and worked on an ESRC funded project on the cognitive neuroscience of social decision making. In 2013, he became a research scientist in the optical and display system division of 3M and in 2016 he joined Oculus Research which then became part of Facebook Reality Labs. James' research focuses on the development of computational models of interactions between humans and their social and physical environments.

**Andrew Duchowski** is a professor of Computer Science at Clemson University. He received his baccalaureate (1990) from Simon Fraser University, Burnaby, Canada, and doctorate (1997) from Texas A&M University, College Station, TX, both in Computer Science. His research and teaching interests include visual attention and perception, eye tracking, computer vision, and computer graphics. He is a noted research leader in the field of eye tracking, having produced a corpus of papers and a monograph related to eye tracking research, and has delivered courses and seminars on the subject at international conferences. He maintains Clemson's eye tracking laboratory, and teaches a regular course on eye tracking methodology attracting students from a variety of disciplines across campus.

### 3 PRE-WORKSHOP PLANS

*Advertisement:* We will broadly advertise the virtual EMICS workshop to researchers and practitioners from academia and industry, and post the call for participation to different communities (e.g. CHI announcements, Vision and CogSci mailing lists) and internal institutional mailing lists. Authors' social media, such as Twitter and Facebook, will be used for announcements. Additionally, we plan to send emails to researchers actively working in related areas to invite them to participate and distribute the announcement. We expect that the 9 invited speakers will bring members of their labs to the workshop. The workshop website will be publicly accessible at <https://emics-2021.github.io/EMICS/>.

*Funding:* We will be securing workshop sponsorship from Facebook Reality Labs and Adobe Inc., from which we already have tentative approval. Upon workshop acceptance, we will also be reaching out to the currently dominant makers of eye tracking technology: SR Research, Tobii and Pupil Labs. Obtained funding will provide the awards for selected best papers to encourage submission and participation to our EMICS workshop.

*Submissions:* We will invite participants to submit a 2 to 4 pages (excluding references) position paper (CHI EA format) focusing on related topics listed in the announcement, to be presented as spotlight presentations at EMICS'21. Submission will be made via email at [emicsCHI21@gmail.com](mailto:emicsCHI21@gmail.com), and peer-reviewed by two external reviewers. Experts from the Cognitive Science and HCI fields will be recruited as reviewers, to capture diverse opinions on the submissions. Acceptance of the submissions will be made based on the quality of the presented work and relevance to the workshop.

#### *Important dates:*

- Call for participation: December 15, 2020
- Submission deadline: February 11, 2021
- Notification of acceptance: February 28, 2021

*Information sharing:* To facilitate the communication and discussion among all workshop attendances, we will set up various channels before, during, and after the workshop:

- a Zotero reference group to share related literature
- a shared Google Drive folder to host all accepted papers and their posters
- a Slack group to identify interesting open questions before the workshop and to continue the discussions and explore further collaborations during and after the meeting.

### 4 VIRTUAL WORKSHOP STRUCTURE

We propose a one day pre-conference virtual workshop at CHI 2021. We aim to bring together 30-40 participants (including organizers and speakers) from academic institutions and industry. The EMICS workshop will feature an invited speaker line-up of 9 experts in Cognitive Science and HCI, two panel discussions with the speakers, and spotlight presentations eliciting a call for abstracts.

The proposed virtual workshop will adapt an asynchronous format similar to the visual conference of SIGGRAPH 2020. All invited talks and accepted papers will be presented through prerecorded videos that are uploaded online. The main workshop will have two sessions, with each of them featuring a panel discussion with a mix of invited speakers from both Cognitive Science and HCI. These two live sessions will take place at two different times such that participants from different time zones can conveniently join at least one.

*Invited talks.* All our invited talks will be presented as pre-recorded videos (of 20 minutes) with captions. We will provide to speakers the instructions on how to use Zoom for auto captioning and how to manually correct the auto-generated scripts. All videos will be uploaded to a YouTube channel, which will be publicly available before the workshop.

*Accepted papers.* Each accept paper will be presented as a pre-recorded video of 5 minutes and they will also be uploaded to the YouTube channel. Similarly, we will provide instructions on how to generate captions for the recorded videos. Additionally, we will ask authors to submit a 40-second video, which will be presented during the spotlight presentation session.

*Session structure.* Each session will be about 2 hours with several breaks in-between. Each session will be hosted on Zoom (with available university licenses capable of hosting 300 participants from the organisers) and moderated by one of the organizers. We will also prepare questions for each speaker as well as questions for the panel discussion to kick-off the conversation. Our Slack group will be used to support the free-form discussions, and we are also considering the usage of other platforms (e.g. Hubs, Gather and Airmee).

*Session 1.* (All time in Eastern Standard Time)

- 09:00 : Welcome & Introduction
- 09:05 : Q&A with invited speakers
  - **Viewing strategies as revealed by eye movements** by Monica Castelhan
  - **Beyond zero order metrics: deep learning based scan-path analytics for cognitive state recognition** by Enkelejda Kasneci
  - **Modeling attention control across visual tasks** by Gregory Zelinsky
  - **Virtual and augmented reality, sensors, eye movements, and their impact on healthcare and security** by Oleg Komogortsev
  - **Eye tracking in program comprehension** by Bonita Sharif
- 09:30 : 5 minutes break
- 09:35 : Panel discussions with all speakers
- 10:25 : 10 minutes break
- 10:35 : Spotlight presentation followed by free-form discussions
- 11:00 : Session closing remarks

#### Session 2. (All time in Eastern Standard Time)

- 17:00 : Welcome & Introduction
- 17:05 : Q&A with invited speakers
  - **Eye tracking measures of cognitive load** by Andrew Duchowski
  - **Using eye movements to assess human comprehension of data visualizations** by Laura Matzen
  - **Gaze sharing for remote collaboration** by Alexandra Papoutsaki
  - **The role of alertness and fatigue in cognition-aware computing** by Tilman Dingler
- 17:25 : 5 minutes break
- 17:30 : Panel discussions with all speakers
- 18:20 : 10 minutes break
- 18:30 : Spotlight presentation followed by free-form discussions
- 18:55 : Session closing remarks

*Confirmed Speakers:* We have already reached out to, and obtained confirmation, from the following proposed speakers.

**Monica Castelhan** is a Professor in Psychology at Queen's University, Canada. Dr. Castelhan studies memory and attentional processes in complex visual environments in real and virtual settings using techniques such as VR, eye tracking and electroencephalogram (EEG). She will give a talk on "Viewing strategies as revealed by eye movements", and explore how strategies as revealed by eye movements have effects on attention, learning, and memory.

**Enkelejda Kasneci** is a Junior Professor of Computer Science at the University of Tübingen, and her main research interests are applied machine learning, eye-tracking technology and applications, and driver assistance systems. Dr. Kasneci will give a talk about "Beyond Zero Order Metrics: Deep Learning based Scanpath Analytics for Cognitive State Recognition".

**Oleg Komogortsev** is currently a Professor of Computer Science at Texas State University. Dr. Komogortsev conducts research

in eye tracking with a focus on health assessment, cyber security (biometrics), bioengineering, human computer interaction, and usability. His talk will be about "Virtual and Augmented Reality, Sensors, Eye Movements, and Their Impact on Healthcare and Security".

**Gregory Zelinsky** is a Professor of Cognitive Science at Stony Brook University. His research attempts to integrate cognitive, computational, and neuroimaging techniques to better understand a broad range of visual cognitive behaviors, including visual search, object representation, working memory, and scene perception. Dr. Zelinsky will talk about "Modeling attention control across visual tasks".

**Bonita Sharif** is an Assistant Professor in the Department of Computer Science and Engineering at University of Nebraska-Lincoln, Lincoln, Nebraska USA. Her research interests are in eye-tracking related to software engineering, emotional awareness, software traceability, and software visualization to support maintenance of large systems. Dr. Sharif will talk about "Eye Tracking in Program Comprehension".

**Andrew Duchowski** is a Professor in the School of Computing at Clemson University and his research and teaching interests include visual attention and perception, eye tracking, computer vision, and computer graphics. Dr. Duchowski is a noted research leader in the field of eye tracking and will talk about "Eye Tracking Measures of Cognitive Load".

**Laura Matzen** is a Principal Member of the Technical Staff in the Cognitive Science and Systems department at Sandia National Laboratories. Her primary research interests lie in using cognitive neuroscience methods to understand how humans process and remember information while performing complex reasoning tasks. Dr. Matzen will talk about "Using eye movements to assess human comprehension of data visualizations".

**Alexandra Papoutsaki** is an Assistant Professor of Computer Science at Pomona College. Her main research focuses on eye tracking and gaze sharing in the context of remote collaboration. She received her PhD from Brown University where she developed a new approach for webcam eye tracking in the browser. Dr. Papoutsaki will talk more about her recent work "Gaze sharing for remote collaboration".

**Tilman Dingler** is a Lecturer in the School of Computing and Information Systems at the University of Melbourne. His research focuses on cognition-aware systems and technologies that support users' information processing capabilities. Dr. Dingler will talk about "The role of alertness and fatigue in cognition-aware computing".

## 5 POST-WORKSHOP PLANS

After the virtual EMICS workshop at CHI, we will summarize the discussions as a report on <https://emics-2021.github.io/EMICS/>. This website will be used to promote the workshop and encourage audiences from diverse disciplines to submit their research. We will also keep all recorded videos available online and release our panel discussions after obtaining everyone's agreement. In the future, we hope that EMICS can grow into a separate venue that can be co-located with related conferences like the ACM Symposium on Eye Tracking Research and Application (ETRA), the annual meeting of the Cognitive Science Society (CogSci) or the Vision Sciences

Society (VSS). We will work towards making EMICS a recognized name that may one day be large enough to be a stand-alone venue.

## 6 CALL FOR PARTICIPATION

The EMICS workshop presents a venue to explore how the connection between eye movements and cognitive processing can be used for applications. We invite papers on topics related to the use of eye movement measures (fixations, saccades, blinks, pupil diameter) for the analysis or prediction of cognitive processing. This could include, but is not limited to, a viewer's cognitive load or affective state, level of expertise, individual traits, task type, task difficulty, as well as the relationship of eye movements to user interfaces broadly construed (driving, reading, problem solving, navigating, data analysis, coding, creating content, etc.).

We invite submissions of a maximum of 4 pages, excluding references. The length of the submission should be commensurate with the contribution. Submissions will be sent out for review to 2 experts representing the Cognitive Science and HCI communities, who will rate the quality of the submission as well as the topic fit to the workshop. Submissions should follow the ACM SIGCHI Extended Abstracts format. Information about where and when to submit the abstracts will be posted on <https://emics-2021.github.io/EMICS/>. At least one author of each accepted paper must attend EMICS and all participants must register for both the workshop and for at least one day of the conference.

Accepted papers will be presented as spotlight presentations at the virtual EMICS workshop. Based on reviewers' recommendation, selected papers will receive best paper awards, funded by the workshop sponsors. Nine invited speakers from both Cognitive Psychology and HCI will present their work, featuring current developments in the relevant fields.

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